

Qatar-2b

R-1794

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_220414 · created 2026-08-03T01:08:04+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459682.4135 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.219 +/- 0.067
Transit depth (Rp/Rs)^2	4.79 +/- 2.92 [%]
Orbital Inclination (inc)	88.63 +/- 1.72
Airmass coefficient 1 (a1)	12.25 +/- 0.3
Airmass coefficient 2 (a2)	-1.38 +/- 0.86
Scatter in the residuals of the lightcurve fit is	2.34 %
Best Comparison Star	#2 - [311, 204]
Optimal Aperture	49.98
Optimal Annulus	148.97
Transit Duration (day)	0.0023 +/- 0.0017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.219 $\pm$ 0.067 vs 0.16327 (deviation 0.83 σ)
Duration fit vs published	0.0023 d vs 0.0767 d (deviation 43.75 σ)
Mid-time O-C	-1928.7 min
Depth SNR	3.3
Residual scatter	2.34 %

Light curve

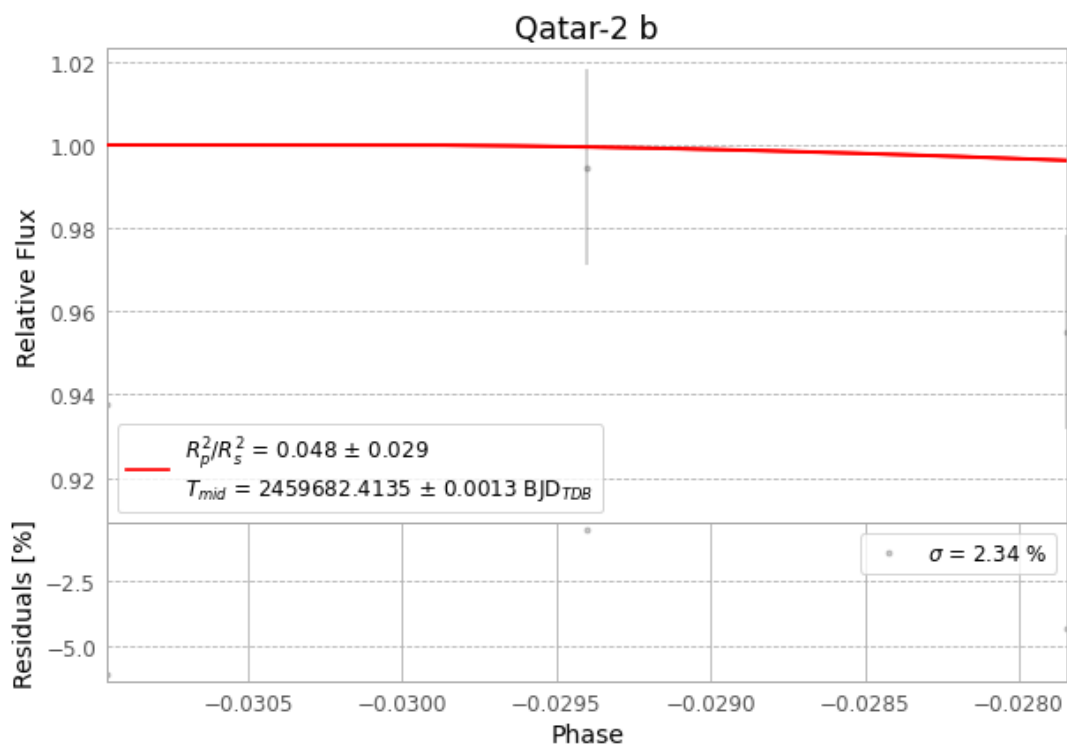


Figure 1 — FinalLightCurve_Qatar-2 b_2022-04-13.png (SHA-256 94b0a3bd7ce92ac0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [344, 103], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2275e7b4c4c2bc59...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

59 FITS frames (39 MB), Qatar-2220414040313.FITS → Qatar-2220414065713.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-220414.log (f3c142b03e3f...), FinalLightCurve_Qatar-2 b_2022-04-13.png (94b0a3bd7ce9...), FinalLightCurve_Qatar-2 b_2022-04-13.csv (354e186ad092...)

Compute resources

Wall time 911.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 01:08Z.